Cisco Router Command Reference

Router-focused IOS commands for the Small Office IT Infrastructure Lab project

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This sheet focuses specifically on router configuration — from basic setup through routing, DHCP, NAT, and access control. Pair it with the switch-side command reference for the full picture of your lab.

1. Router Command Modes

Command	*What it does*
`Router>`	User EXEC — default mode after login, very limited
`Router#`	Privileged EXEC — enter with: enable
`Router(config)#`	Global configuration — enter with: configure terminal
`Router(config-if)#`	Interface configuration — enter with: interface gig0/0
`Router(config-subif)#`	Sub-interface configuration — enter with: interface gig0/0.10
`Router(config-router)#`	Routing protocol configuration — enter with: router ospf 1 (or rip, eigrp, etc.)

2. Basic Router Setup

Command	*What it does*
`hostname R1`	Sets the device name shown in the prompt
`enable secret cisco123`	Sets an encrypted password for privileged mode
`line console 0 / password cisco / login`	Requires a password on the console port
`line vty 0 4 / password cisco / login`	Requires a password for Telnet/SSH sessions
`service password-encryption`	Encrypts plain-text passwords stored in the config
`no ip domain-lookup`	Stops the router trying to DNS-resolve mistyped commands
`banner motd # Authorized access only #`	Sets a login banner

3. Interface Configuration Basics

Command	*What it does*
`interface gig0/0`	Selects a physical interface to configure
`description Link to Switch0`	Adds a human-readable label to the interface (shows in show run)
`ip address 192.168.1.1 255.255.255.0`	Assigns an IP address and subnet mask
`no shutdown`	Activates the interface — interfaces are administratively down by default
`shutdown`	Manually disables an interface
`show ip interface brief`	Quick view of every interface's IP and up/down status
`show interfaces gig0/0`	Detailed stats, errors, and status for one interface

Interface naming varies by router model: FastEthernet0/0, GigabitEthernet0/0, and Serial0/0/0 are all common — use `show ip interface brief` to see exactly what your router has.

4. Sub-Interfaces & Router-on-a-Stick

Used when one physical router port needs to serve multiple VLANs coming from a trunk link.

Command	*What it does*
`interface gig0/0.10`	Creates sub-interface .10 under the physical interface (number is arbitrary, but match the VLAN for clarity)
`encapsulation dot1Q 10`	Tags this sub-interface to VLAN 10 — must match the switch trunk's VLAN
`encapsulation dot1Q 99 native`	Marks VLAN 99 as the native (untagged) VLAN, if your trunk uses one
`ip address 192.168.10.1 255.255.255.0`	Sets the gateway IP that VLAN 10 devices will use
`no shutdown (on the physical interface)`	The physical interface must be up for any of its sub-interfaces to work

5. Static & Default Routing

Command	*What it does*
`ip route 192.168.50.0 255.255.255.0 10.0.0.2`	Manually adds a route: to reach this network, send traffic to this next-hop IP
`ip route 0.0.0.0 0.0.0.0 203.0.113.1`	Default route — "anything not otherwise known, send here" (typically your ISP/internet edge)

Command	*What it does*
`show ip route`	Displays the full routing table, including static, connected, and learned routes
`no ip route 192.168.50.0 255.255.255.0 10.0.0.2`	Removes a specific static route

6. Dynamic Routing Basics (Optional)

For labs with more than one router. Not usually needed for a single-router small-office design, but useful to recognize.

Command	*What it does*
`router ospf 1`	Starts OSPF process ID 1 and enters routing config mode
`network 192.168.10.0 0.0.0.255 area 0`	Advertises this network into OSPF area 0
`router rip / version 2 / network 192.168.10.0`	Simpler alternative to OSPF for small/legacy labs
`show ip protocols`	Shows which routing protocols are active and their settings
`show ip ospf neighbor`	Lists OSPF neighbors the router has formed adjacencies with

7. DHCP Server Configuration

Command	*What it does*
`ip dhcp excluded-address 192.168.10.1 192.168.10.10`	Reserves addresses DHCP should never hand out
`ip dhcp pool SALES`	Creates a DHCP pool named SALES
`network 192.168.10.0 255.255.255.0`	Sets the subnet this pool serves (inside the pool)
`default-router 192.168.10.1`	Tells clients their gateway address
`dns-server 8.8.8.8`	Tells clients which DNS server to use
`lease 7`	Sets the DHCP lease time to 7 days (default is 1 day)
`show ip dhcp binding`	Lists which devices have leased which IP addresses
`show ip dhcp pool`	Shows pool stats: total addresses, leased, available

8. NAT / PAT for Internet Access

Needed if your office router connects to a simulated ISP/internet and internal private addresses (192.168.x.x) need to reach it.

Command	*What it does*
`interface gig0/1 / ip nat outside`	Marks the internet-facing interface as the NAT outside interface
`interface gig0/0.10 / ip nat inside`	Marks an internal VLAN sub-interface as NAT inside (repeat per VLAN)
`access-list 1 permit 192.168.10.0 0.0.0.255`	Defines which internal addresses are allowed to be translated
`ip nat inside source list 1 interface gig0/1 overload`	Enables PAT ("NAT overload") so all listed internal hosts share the outside interface's IP
`show ip nat translations`	Shows active NAT/PAT translations
`show ip nat statistics`	Shows NAT hit counts and configuration summary

9. Access Control Lists (ACLs)

Command	*What it does*
`access-list 10 permit 192.168.10.0 0.0.0.255`	Standard ACL: matches by source address only (wildcard mask, not subnet mask)
`access-list 101 deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255`	Extended ACL: matches by source AND destination (more precise)
`access-list 101 permit ip any any`	Required catch-all — ACLs deny everything else by default
`interface gig0/0.10 / ip access-group 101 in`	Applies the ACL to traffic entering this interface
`show access-lists`	Shows all ACLs and how many packets matched each line

10. Verification & Troubleshooting

Command	*What it does*
`show running-config`	Shows the active configuration currently in memory
`show startup-config`	Shows the configuration saved for next reboot
`show ip interface brief`	Quick status/IP check across all interfaces
`show ip route`	Displays the routing table
`ping 192.168.20.1`	Tests basic reachability to an address

Command	*What it does*
`traceroute 192.168.20.1`	Shows the hop-by-hop path to a destination
`show cdp neighbors`	Lists directly connected Cisco devices
`show version`	Shows IOS version, uptime, and hardware info

11. Saving & Backing Up Configuration

Command	*What it does*
`copy running-config startup-config`	Saves your current config so it survives a reboot (or: write memory)
`copy running-config tftp`	Backs up the config to a TFTP server
`erase startup-config`	Wipes the saved config (router reverts to blank setup on next reboot)
`reload`	Restarts the router — will prompt to save first if there are unsaved changes

Save often. Losing power or closing Packet Tracer without saving means starting the router config over from scratch.

12. One-Line Cheat Sheet

Command	*What it does*
`enable`	Enter privileged mode
`configure terminal`	Enter global config mode
`interface`	Configure a specific interface
`ip address`	Assign an IP to an interface
`no shutdown`	Turn an interface on
`encapsulation dot1Q`	Tag a sub-interface to a VLAN
`ip route 0.0.0.0 0.0.0.0`	Set a default route
`write memory`	Save the configuration

Reference sheet prepared for the Small Office IT Infrastructure Lab project. Keep this handy while configuring Router0 in Cisco Packet Tracer.